Software Requirements Specification (SRS)

Traffic Management System (Simulation)

Problem Statement No. 16

Project: Traffic Management System (Simulation)

Version: 1.0

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Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Traffic Management System (Simulation), developed for Problem Statement No. 16. It defines the functional and non-functional requirements, external interfaces, and verification criteria for a system that manages live traffic signal operations and provides a simulation environment for planning and analysing traffic flow.

1.2 Scope

The system covers: real-time traffic signal control at intersections based on sensor/camera input; vehicle and pedestrian detection; a simulation engine for modelling road networks and evaluating signal timing plans offline; incident detection and alerting; and reporting/analytics for traffic operations. It excludes physical construction of road infrastructure, vehicle-side (in-car) systems, and core city-planning/zoning decisions beyond providing data to support them.

1.3 Audience

Developers, QA Engineers, System Integrators, Traffic Engineers/Analysts, Traffic Operations staff, and Assessment Evaluators.

1.4 Definitions

List of acronyms: TMS (Traffic Management System), UI (User Interface), API (Application Programming Interface), GUI (Graphical User Interface), IoT (Internet of Things), RBAC (Role-Based Access Control), MFA (Multi-Factor Authentication), NFR (Non-Functional Requirement), FR (Functional Requirement), RTO/RPO (Recovery Time/Point Objective), WCAG (Web Content Accessibility Guidelines).

1.5 References

- IEEE 830 / ISO/IEC/IEEE 29148 - Software Requirements Specification guidelines
- WCAG 2.1 - Web Content Accessibility Guidelines
- Local traffic-control device and road-safety standards (jurisdiction-specific)
- Applicable data-protection regulation for captured imagery/PII

1.6 Document conventions

Requirement IDs use the prefixes TMS-F-### for functional requirements, TMS-NF-### for non-functional requirements, and PRJ-SR-### for security requirements. The keyword "shall" indicates a mandatory requirement; "should" indicates a recommended but non-mandatory capability. Priority values are High, Medium, or Low.

2. Overall description

2.1 Product perspective

The Traffic Management System is a networked platform combining field hardware (signal controllers, vehicle/pedestrian sensors, cameras) at intersections with a central software platform used by operators and analysts. It also provides an offline/what-if Simulation Engine that models road networks and signal plans without requiring live field hardware, enabling planning and training before deployment.

2.2 Major product functions (detailed)

- Real-time adaptive and manual traffic signal control
- Vehicle detection, classification, and counting
- Pedestrian detection and crossing management

- Emergency-vehicle signal preemption
- Road-network modelling and traffic-flow simulation
- Simulation scenario comparison and replay
- Incident detection and operator alerting
- Reporting and analytics (volume, incidents, performance)
- Administration: role-based access, intersection configuration

2.3 User roles and characteristics (expanded)

- Traffic Operator: monitors live signal status, handles manual overrides and incident response; expects a clear, low-latency console.
- Traffic Engineer/Analyst: builds and runs simulation scenarios, analyses performance metrics and reports.
- System Administrator: configures intersections, manages users/roles, and maintains system health.
- Emergency Services: triggers preemption requests and receives incident notifications.
- General Public/Commuter (indirect): benefits from improved signal timing; not a direct system user.

2.4 Operating environment

Field controllers run an embedded real-time OS at each intersection, connected over a secure private or cellular WAN to a central control platform hosted on-premises or in the cloud. The control centre and simulation environment run on standard server/desktop operating systems, accessed via web-based UI. Field cabinets require UPS backup power and must tolerate outdoor environmental conditions (temperature, humidity, vibration).

2.5 Constraints

Compliance with local traffic-control device standards and road-safety regulations; data-protection requirements for captured video/imagery (e.g., number-plate data); use of approved cryptographic modules for field-to-centre communication; dependency on signal-controller and sensor vendor APIs; simulation results are estimates and do not replace formal traffic-engineering sign-off.

2.6 Assumptions and dependencies

- Field intersections have reliable power (with UPS backup) and network connectivity for the majority of operating hours.
- Sensor and camera hardware meets the accuracy specifications assumed by the detection/classification requirements.
- Signal controller vendors provide a documented API sufficient to implement TMS-F-001, TMS-F-002, and TMS-F-004.
- Simulation scenarios are built and reviewed by users with basic traffic-engineering knowledge.
- Emergency Services systems used for preemption requests support the authentication mechanism required by PRJ-SR-002 or an equivalent secure channel.

2.7 Apportioning of requirements

Requirements marked Medium or Low priority (e.g., TMS-F-012 scenario comparison, TMS-F-013 replay, TMS-NF-008 portability) may be deferred to a later release phase if project timelines require, subject to stakeholder agreement. All High-priority functional and security requirements are considered mandatory for the initial release.

3. External interface requirements

3.1 User interfaces

Primary UI: web-based operator console and analyst/simulation workbench with a clear menu structure and role-appropriate dashboards. Accessibility: screen-reader friendly prompts and high-contrast mode conforming to WCAG 2.1 AA where applicable.

3.2 Hardware interfaces

- Traffic signal controller (vendor-specific field interface)
- Inductive-loop / radar / camera vehicle detectors
- Pedestrian push-button and crosswalk sensors
- CCTV/traffic cameras for classification and incident detection
- Roadside unit / cabinet status and operator panel

3.3 Software interfaces

- Signal Controller API (authorised messages over TLS) for phase/timing commands
- Detection/Analytics API for vehicle and pedestrian sensor data ingestion
- Emergency Services notification/preemption interface
- Monitoring & Logging API for health checks and alerts
- Optional GIS/mapping service for road-network import

3.4 Communications

TLS 1.2+ enforced for all field-to-centre and centre-to-external-service communication, with mutual TLS for signal-control commands where required. Retry/backoff strategy with local queuing at the field controller during network outage, with secure resynchronisation on reconnect.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow TMS-F-###.

4.1 Traffic Signal Control

Description: Control and coordinate signal timing at intersections using real-time sensor data, manual operator input, and emergency preemption requests.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
TMS-F-001	The system shall compute and apply signal phase timing at each intersection based on real-time vehicle density data received from detection sensors.	Functional	High	Traffic Engineering	AC-TMS-F-001: Given sensor density input, computed green time changes within one control cycle. Test: TC-Sig-01	Requires VDC module
TMS-F-002	The system shall allow an authorised Traffic Operator to manually override active signal	Functional	High	Operations	AC-TMS-F-002: Manual override applied within 2s and reverts to auto mode on	Requires RBAC

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
	timing via the control console.				operator command. Test: TC-Sig-02	
TMS-F-003	The system shall record every signal phase/state change with a timestamp, intersection ID, and triggering source (auto/manual/preemption) to an audit log.	Functional	High	Audit/Compliance	AC-TMS-F-003: 100% of state changes present in audit log during test window. Test: TC-Sig-03	Feeds NFR audit retention
TMS-F-004	The system shall support emergency-vehicle signal preemption, turning the approach corridor green upon a validated preemption request.	Functional	High	Emergency Services	AC-TMS-F-004: Preemption applied within 3s of validated request; normal operation resumes after clearance. Test: TC-Sig-04	Requires secure preemption channel
TMS-F-005	The system shall support green-wave coordination, synchronising signal offsets across adjacent intersections along a configured corridor.	Functional	Medium	Traffic Engineering	AC-TMS-F-005: Simulated platoon crosses 3 coordinated intersections without a full stop at target speed. Test: TC-Sig-05	Depends on corridor config

4.2 Vehicle & Pedestrian Detection

Description: Detect, classify, and count vehicles and pedestrians at each monitored approach to drive signal decisions and analytics.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
TMS-F-006	The system shall detect and classify vehicles by type (car, bus, truck, two-wheeler) using camera/sensor feeds at each monitored approach.	Functional	High	Operations	AC-TMS-F-006: Classification accuracy ≥ 90% on validation dataset. Test: TC-VDC-01	Requires camera/AI model
TMS-F-007	The system shall count vehicle volume per lane for each configurable time interval and persist counts for analytics.	Functional	High	Analytics	AC-TMS-F-007: Counts within ±5% of manual ground-truth count over a 15-minute sample. Test: TC-VDC-02	Feeds Reporting module
TMS-F-008	The system shall detect pedestrian presence at a crosswalk push-button/sensor and trigger the	Functional	High	Public Safety	AC-TMS-F-008: Pedestrian phase triggered within one cycle of a valid request. Test: TC-VDC-03	Requires crosswalk sensor

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
	pedestrian signal phase within the next safe cycle.					

4.3 Simulation Engine

Description: Model road networks and simulate traffic flow to evaluate signal timing plans and operational scenarios offline.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
TMS-F-009	The system shall simulate traffic flow on a defined road network given input parameters such as vehicle arrival rate and a signal timing plan.	Functional	High	Traffic Engineer/Analyst	AC-TMS-F-009: Simulation completes and produces vehicle trajectories for a 1-hour scenario. Test: TC-Sim-01	Core simulation engine
TMS-F-010	The system shall allow a user to create or import a road-network topology (intersections, lanes, signal phases) for use in a simulation scenario.	Functional	High	Traffic Engineer/Analyst	AC-TMS-F-010: Imported network validated and renders correctly with no orphan links. Test: TC-Sim-02	Supports common map formats
TMS-F-011	The system shall compute simulation performance metrics, including average delay, queue length, and throughput, for each completed run.	Functional	High	Traffic Engineer/Analyst	AC-TMS-F-011: Metrics computed and displayed within 5s of run completion. Test: TC-Sim-03	Feeds comparison feature
TMS-F-012	The system shall allow a user to compare metrics from two or more simulation scenarios side-by-side.	Functional	Medium	Traffic Engineer/Analyst	AC-TMS-F-012: Comparison view renders selected scenarios with aligned metrics. Test: TC-Sim-04	Depends on TMS-F-011
TMS-F-013	The system shall allow playback/replay of a completed simulation run at variable speed (0.5x-8x).	Functional	Medium	Traffic Engineer/Analyst	AC-TMS-F-013: Replay controls change playback speed without desync of vehicle positions. Test: TC-Sim-05	UI/rendering dependency

4.4 Incident Detection & Alerts

Description: Detect probable incidents from sensor/camera data and notify operators and emergency services.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
TMS-F-014	The system shall detect probable traffic incidents (congestion build-up, stalled vehicle, collision) using anomaly detection on sensor/camera data.	Functional	High	Operations/Public Safety	AC-TMS-F-014: Injected test-incident scenario detected within 60s. Test: TC-Inc-01	Requires VDC + analytics
TMS-F-015	The system shall generate a real-time alert to on-duty Traffic Operators, including location, type, and severity, when an incident is detected.	Functional	High	Operations	AC-TMS-F-015: Alert appears on operator console within 5s of detection. Test: TC-Inc-02	Depends on TMS-F-014
TMS-F-016	The system shall allow an operator to dispatch or notify emergency services directly from the incident alert screen.	Functional	High	Emergency Services	AC-TMS-F-016: Notification transmitted and acknowledged; action logged. Test: TC-Inc-03	Requires external service interface

4.5 Reporting, Analytics & Administration

Description: Generate operational reports and provide role-based administration of system configuration.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
TMS-F-017	The system shall generate daily, weekly, and monthly traffic reports covering volume, incidents, and signal performance per intersection.	Functional	High	Analytics/Management	AC-TMS-F-017: Scheduled reports generated automatically and available for download. Test: TC-Rpt-01	Depends on TMS-F-007, 011
TMS-F-018	The system shall enforce role-based access control (RBAC) distinguishing Operator, Administrator, and Analyst roles and their permitted actions.	Functional	High	Security	AC-TMS-F-018: User with Analyst role denied access to signal-override function. Test: TC-Adm-01	Security dependency
TMS-F-019	The system shall allow an Administrator to configure intersection parameters and default signal timing plans	Functional	High	Operations	AC-TMS-F-019: Configuration change is validated, versioned, and applied without restart. Test: TC-Adm-02	Requires audit trail (TMS-F-003)

Req ID	Requirement (shall…)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments/Dependencies
	through an administration UI.					

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs TMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TMS-NF-001	Signal control decisions (sensor input to phase update) shall complete within 1 second for 95% of control cycles under normal load.	Performance	High	95th percentile ≤ 1s in production-like load test. Test: TC-Perf-01
TMS-NF-002	The system shall provide 99.5% availability per month for the signal-control service, excluding scheduled maintenance windows.	Reliability	High	Uptime reports show ≥99.5% per month. Test: Ops reports
TMS-NF-003	All video/image and command data in transit and at rest must be protected per applicable data-protection requirements; no plaintext storage of identifiable footage.	Security/Compliance	High	Security audit checklist pass. Test: TC-Sec-01
TMS-NF-004	The operator console and public-facing dashboards shall conform to WCAG 2.1 AA accessibility guidelines where applicable.	Usability/Accessibility	Medium	Accessibility audit pass. Test: TC-UX-01
TMS-NF-005	The system shall support simulation and live monitoring of road networks scaling up to 500 intersections without exceeding defined response-time thresholds.	Scalability	Medium	Load test at 500-intersection scale meets NFR-001 thresholds. Test: TC-Scale-01
TMS-NF-006	Field controller software shall continue essential signal operation for at least 4 hours during a total loss of connectivity to the central platform, using the last known-good timing plan.	Reliability/Resilience	High	Simulated WAN outage for 4 hours; signals continue safe default operation. Test: TC-Rel-01
TMS-NF-007	The system shall be maintainable such that a new signal-timing algorithm module can be added without modifying the core detection or reporting modules.	Maintainability	Medium	Module added via defined plugin interface with no changes outside module boundary. Test: TC-Maint-01
TMS-NF-008	The central platform shall be deployable on at least two different cloud/on-prem environments without code changes, using externalised configuration.	Portability	Medium	Successful deployment on two target environments from the same build artifact. Test: TC-Port-01
TMS-NF-009	The system shall provide daily automated backups of configuration and historical traffic data, with a Recovery Point Objective (RPO) of 24 hours and Recovery Time Objective (RTO) of 4 hours.	Backup/Recovery	High	Restore drill completes within RTO using a backup no older than RPO. Test: TC-BCP-01
TMS-NF-010	The system shall interoperate with third-party GIS and city dashboard platforms via a documented, versioned REST API.	Interoperability	Medium	Sample third-party client consumes API per published contract without errors. Test: TC-Interop-01

5.1 Security

5.1.1 Security Objectives

- SO-1: Ensure the integrity and confidentiality of traffic-control commands to prevent unauthorised signal manipulation.
- SO-2: Protect captured video/image data and personally identifiable information (e.g., number plates) in compliance with applicable data-protection regulations.
- SO-3: Ensure the availability of the signal-control system against denial-of-service and cyber-attacks that could impact public safety.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
PRJ-SR-001	TLS 1.2+ shall be mandatory for all network connections between field devices, the control centre, and external services.	Security	High	Protocol scan shows no connection below TLS 1.2. Test: TC-Sec-02
PRJ-SR-002	All operator actions that change signal control state shall require multi-factor authentication (MFA).	Security	High	Login without second factor is rejected. Test: TC-Sec-03
PRJ-SR-003	Video/image data containing number plates or other identifiable information shall be encrypted at rest and all access shall be logged.	Security	High	Access log entry created for every retrieval; storage encryption verified. Test: TC-Sec-04
PRJ-SR-004	The system shall implement role-based access control with least-privilege defaults for all administrative functions.	Security	High	Attempted privilege escalation by a low-privilege role is denied and logged. Test: TC-Sec-05
PRJ-SR-005	The system shall detect and alert on anomalous command patterns (e.g., rapid repeated overrides, off-hours admin access) indicative of unauthorised access.	Security	Medium	Simulated anomalous pattern triggers a security alert within 60s. Test: TC-Sec-06

5.1.3 Risk and threat summary

The table below summarises key risks considered during requirements analysis and their mitigations, cross-referenced to the relevant requirements.

Risk ID	Threat / Risk	Likelihood	Impact	Mitigation
RSK-001	Unauthorised remote access to a signal controller leading to unsafe phase changes.	Medium	High	Mutual TLS, MFA for admin/operator actions, network segmentation, anomaly detection (PRJ-SR-005).
RSK-002	Loss of WAN connectivity between field controller and control centre.	Medium	Medium	Local fail-safe timing plan, offline queuing, automatic resync (TMS-NF-006).
RSK-003	Inaccurate vehicle/pedestrian detection leading to poor signal decisions.	Medium	Medium	Sensor redundancy, periodic calibration, classification accuracy monitoring (TMS-F-006).
RSK-004	Sensitive video/PII data exposure (e.g., number plates) through unauthorised access or breach.	Low	High	Encryption at rest/in transit, access logging, least-privilege RBAC (PRJ-SR-003, PRJ-SR-004).
RSK-005	Simulation results misinterpreted as guaranteed real-world outcomes.	Medium	Low	Clear labelling of simulation outputs as estimates; sign-off by qualified traffic engineer required before field deployment.
RSK-006	Scalability bottlenecks as the number of managed intersections grows.	Medium	Medium	Horizontal scaling design, load testing at target scale (TMS-NF-005), capacity monitoring.

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.

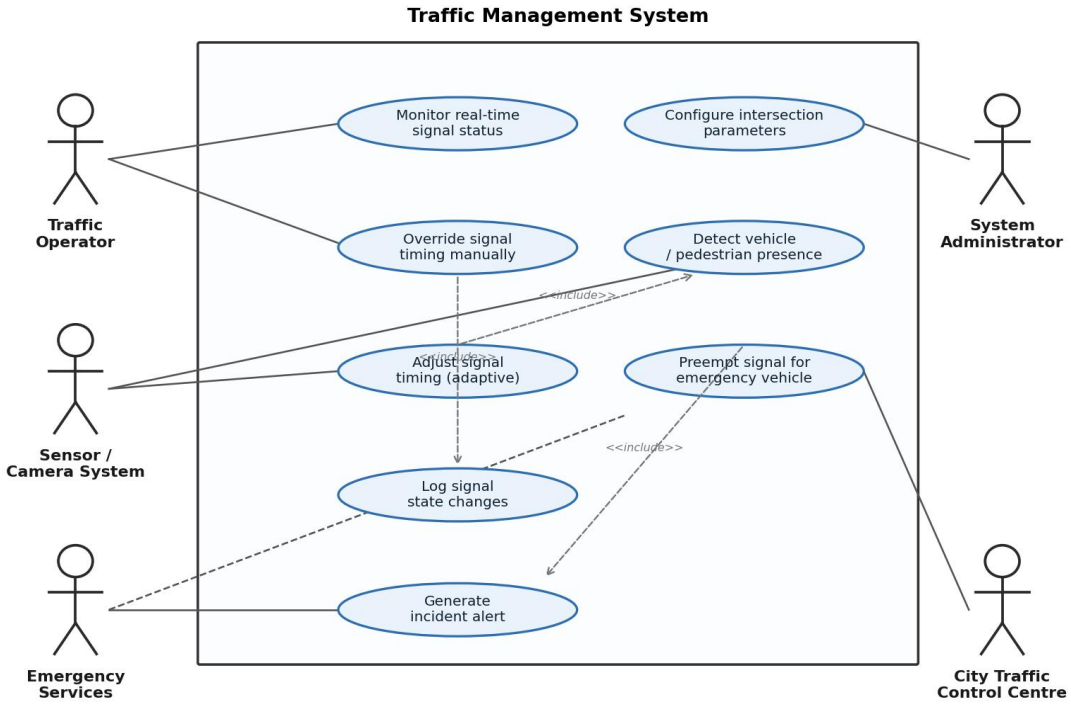
- Acceptance test suites: Signal Control, Detection, Simulation, Incident Management, Reporting, Security, and Accessibility tests.

7. System models and diagrams

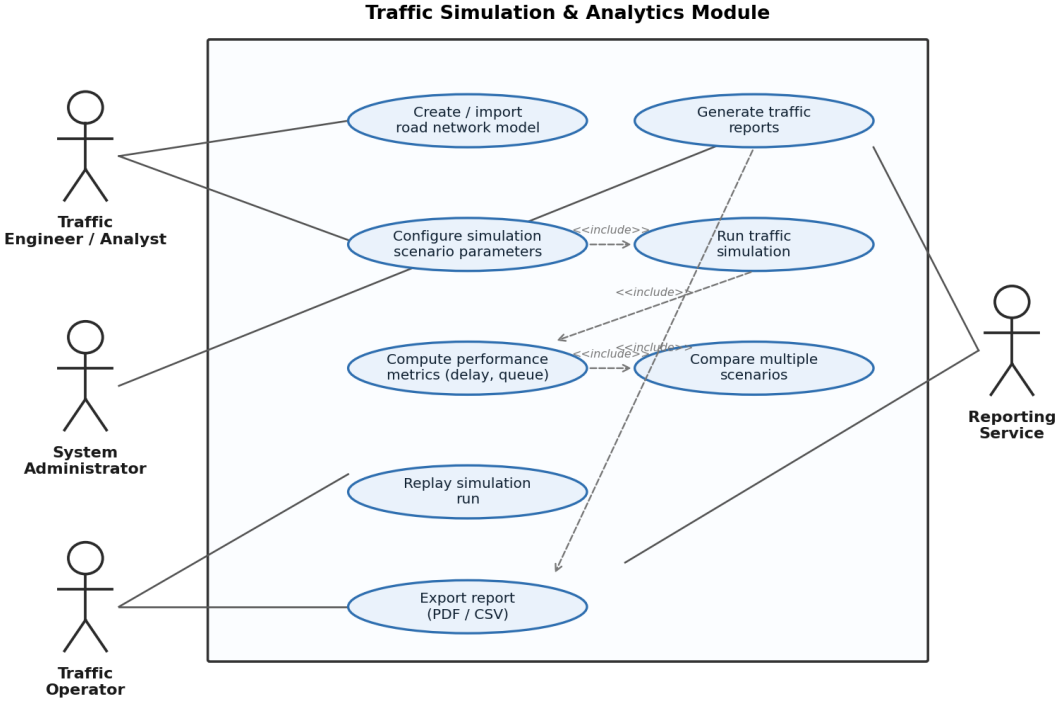
7.1 UML Use-Case diagrams

Two use-case diagrams are provided below, covering (a) live traffic signal control and monitoring, and (b) the simulation and analytics module.

7.1.1 Traffic Signal Control & Monitoring



7.1.2 Traffic Simulation & Analytics


Traffic Simulation & Analytics Module
Traffic Engineer / Analyst
System Administrator
Traffic Operator
Reporting Service
Create / import road network model
Generate traffic reports
Configure simulation scenario parameters
Run traffic simulation
Compute performance metrics (delay, queue)
Compare multiple scenarios
Replay simulation run
Export report (PDF / CSV)
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7.2 Use-case specifications

The following textual use-case specifications elaborate the primary use cases shown in the diagrams above, in line with the functional requirements in Section 4.

UC-01 Adaptive Signal Timing Update

Actors: Sensor/Camera System (primary), Traffic Operator (monitors)

Preconditions: Intersection is online and reporting to the central platform; adaptive mode is enabled for the intersection.

Main flow:

1. Sensor/Camera System reports current vehicle density per approach to the Signal Control Module.
2. Signal Control Module computes an updated phase timing plan for the intersection.
3. System validates the new plan against configured safety minimums/maximums.
4. System applies the new phase timing at the next safe transition point.
5. System logs the change to the audit log (TMS-F-003).

Alternate / exception flow:

- If validation fails, the system retains the previous timing plan and raises a configuration warning to the Administrator.
- If the intersection is in manual-override mode, the computed plan is stored but not applied until override is cleared.

Postconditions: Intersection operates on the updated (or retained) timing plan; the change is recorded in the audit log.

UC-02 Manual Signal Override

Actors: Traffic Operator

Preconditions: Operator is authenticated with the Operator role for the target intersection.

Main flow:

1. Operator selects an intersection on the control console.
2. Operator selects a manual phase and duration, or requests an all-red hold.
3. System validates the request against safety interlocks (e.g., minimum pedestrian clearance).
4. System applies the manual command and suppresses adaptive updates for that intersection.
5. System logs the override action, operator ID, and timestamp.

Alternate / exception flow:

- If the request violates a safety interlock, the system rejects it and displays the reason.
- Operator may return the intersection to adaptive mode at any time, ending the override.

Postconditions: Intersection reflects the operator's manual command until override is cleared; action is auditable.

UC-03 Emergency Vehicle Preemption

Actors: Emergency Services (primary), Traffic Operator (monitors)

Preconditions: Emergency vehicle is equipped with an authorised preemption transmitter or dispatch-linked request; corridor is configured for preemption.

Main flow:

1. Emergency Services system sends a validated preemption request for a corridor.
2. System authenticates the request and identifies affected intersections.
3. System transitions affected intersections to green along the emergency route ahead of the vehicle.
4. System notifies the Traffic Operator that preemption is active.
5. On request clearance or timeout, the system restores normal/adaptive operation.

Alternate / exception flow:

- If the request cannot be authenticated, it is rejected and logged as a security event (PRJ-SR-005).
- If two preemption requests conflict at a shared intersection, priority rules resolve the conflict and both requesters are notified.

Postconditions: Emergency corridor received priority signal access; all actions are logged for audit.

UC-04 Build and Run a Simulation Scenario

Actors: Traffic Engineer/Analyst

Preconditions: Analyst is authenticated; a road-network model exists or can be imported.

Main flow:

1. Analyst creates a new scenario and imports or selects a road-network topology.
2. Analyst configures input parameters: vehicle arrival rates, signal timing plan, simulation duration.
3. Analyst starts the simulation run.
4. Simulation Engine executes the run and computes performance metrics (delay, queue length, throughput).
5. System displays results and stores the run for later replay or comparison.

Alternate / exception flow:

- If the network model contains errors (e.g., orphan links), the system flags them before allowing the run to start.
- Analyst may cancel a long-running simulation; partial results are discarded.

Postconditions: A completed simulation run with computed metrics is available for review, comparison, and replay.

UC-05 Incident Detection and Response

Actors: Sensor/Camera System (primary), Traffic Operator, Emergency Services

Preconditions: Incident-detection analytics are enabled for the monitored approach.

Main flow:

1. Sensor/Camera System streams data to the Incident Detection module.
2. System applies anomaly detection and flags a probable incident with location, type, and severity.
3. System raises a real-time alert on the Traffic Operator console.
4. Operator reviews the alert and, if warranted, dispatches or notifies Emergency Services directly from the alert screen.
5. Operator marks the incident as acknowledged/resolved once cleared.

Alternate / exception flow:

- If the Operator determines the alert is a false positive, it is dismissed and logged to improve future detection tuning.
- If no Operator acknowledges the alert within a configured timeout, the system escalates it (e.g., secondary notification).

Postconditions: Incident is tracked from detection through resolution, with all actions time-stamped for reporting.

UC-06 Generate and Export a Traffic Report

Actors: Traffic Engineer/Analyst, System Administrator

Preconditions: Sufficient volume/incident/performance data exists for the requested reporting period.

Main flow:

1. User selects a report type (volume, incident, or signal performance) and a reporting period.
2. System aggregates the underlying data for the selected intersections and period.

3. System renders the report on-screen with summary charts and tables.
4. User exports the report as PDF or CSV.

Alternate / exception flow:

- If no data exists for the selected period, the system informs the user rather than generating an empty report.
- Reports may also be scheduled to generate and distribute automatically (TMS-F-017).

Postconditions: A report is available on-screen and/or as an exported file for distribution.

8. Requirements Traceability Matrix (RTM)

The RTM below traces representative functional, non-functional, and security requirements to design references, implementing modules, and test cases. The full set of requirement IDs (TMS-F-001 to TMS-F-019, TMS-NF-001 to TMS-NF-010, PRJ-SR-001 to PRJ-SR-005) should be traced in the same manner as the project proceeds to design and test planning.

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TMS-F-001	Adaptive signal timing	4.1 / DS-Sig-01	SignalControlModule	TC-Sig-01	N	
TMS-F-002	Manual override	4.1 / DS-Sig-02	SignalControlModule	TC-Sig-02	N	
TMS-F-003	Audit logging of state changes	4.1 / DS-Sig-03	SignalControlModule	TC-Sig-03	N	
TMS-F-004	Emergency preemption	4.1 / DS-Sig-04	SignalControlModule	TC-Sig-04	N	
TMS-F-005	Green-wave coordination	4.1 / DS-Sig-05	SignalControlModule	TC-Sig-05	N	
TMS-F-006	Vehicle classification	4.2 / DS-VDC-01	DetectionModule	TC-VDC-01	N	
TMS-F-007	Vehicle counting	4.2 / DS-VDC-02	DetectionModule	TC-VDC-02	N	
TMS-F-008	Pedestrian detection	4.2 / DS-VDC-03	DetectionModule	TC-VDC-03	N	
TMS-F-009	Traffic flow simulation	4.3 / DS-Sim-01	SimulationEngine	TC-Sim-01	N	
TMS-F-010	Road-network import	4.3 / DS-Sim-02	SimulationEngine	TC-Sim-02	N	
TMS-F-011	Simulation performance metrics	4.3 / DS-Sim-03	SimulationEngine	TC-Sim-03	N	
TMS-F-012	Scenario comparison	4.3 / DS-Sim-04	SimulationEngine	TC-Sim-04	N	
TMS-F-013	Simulation replay	4.3 / DS-Sim-05	SimulationEngine	TC-Sim-05	N	
TMS-F-014	Incident detection	4.4 / DS-Inc-01	IncidentModule	TC-Inc-01	N	
TMS-F-015	Operator alerting	4.4 / DS-Inc-02	IncidentModule	TC-Inc-02	N	
TMS-F-016	Emergency dispatch notification	4.4 / DS-Inc-03	IncidentModule	TC-Inc-03	N	
TMS-F-017	Traffic reports	4.5 / DS-Rpt-01	ReportingModule	TC-Rpt-01	N	
TMS-F-018	Role-based access control	4.5 / DS-Adm-01	AdminModule	TC-Adm-01	N	
TMS-F-019	Intersection configuration	4.5 / DS-Adm-02	AdminModule	TC-Adm-02	N	
TMS-NF-001	Control loop latency target	5 / DS-Perf-01	SignalControlModule	TC-Perf-01	N	
TMS-NF-002	Monthly availability target	5 / DS-Rel-01	Platform/Infra	Ops reports	N	
TMS-NF-006	Offline fail-safe operation	5 / DS-Rel-02	SignalControlModule	TC-Rel-01	N	
TMS-NF-009	Backup and recovery	5 / DS-BCP-01	Platform/Infra	TC-BCP-01	N	

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
PRJ-SR-001	TLS 1.2+ mandatory	5.1.2 / DS-Sec-01	Platform/Infra	TC-Sec-02	N	
PRJ-SR-002	MFA for control actions	5.1.2 / DS-Sec-02	AdminModule	TC-Sec-03	N	
PRJ-SR-003	Encryption of PII video data	5.1.2 / DS-Sec-03	DetectionModule	TC-Sec-04	N	

9. Appendix

9.1 Glossary

Term	Definition
Adaptive signal control	Signal timing that adjusts automatically based on real-time traffic conditions rather than a fixed schedule.
Green-wave coordination	Synchronising signal offsets along a corridor so that vehicles travelling at a target speed encounter consecutive green lights.
Preemption	Overriding normal signal operation to give priority right-of-way, typically to emergency vehicles.
Simulation scenario	A configured set of road-network, demand, and signal-timing inputs used to run a traffic simulation.
Queue length	The number of vehicles waiting at an intersection approach at a given time, used as a performance metric.
RTM	Requirements Traceability Matrix - maps requirements to design, implementation, and test artefacts.

9.2 Open issues

- Final selection of signal-controller hardware vendor and confirmation of API capabilities (affects TMS-F-001, TMS-F-002, TMS-F-004).
- Confirmation of applicable data-protection regulation for number-plate/video data in the target deployment jurisdiction.
- Target scale (number of intersections) for the first production deployment, to validate TMS-NF-005 thresholds.

9.3 Future enhancements (out of current scope)

- Predictive traffic-demand forecasting using historical and weather data.
- Inter-agency data sharing with neighbouring city traffic authorities.
- Connected-vehicle (V2X) integration for direct vehicle-to-infrastructure signal requests.
- Mobile app for public real-time traffic and incident information.